

CMS Mathematical Modelling Competition

2026

Please read the [Competition Handbook](#) before attempting the problems

Timing Selection for NIPT and Determination of Foetal Abnormalities

NIPT (Non-invasive Prenatal Testing) is a prenatal screening technique that analyses cell-free foetal DNA fragments obtained from maternal blood samples to determine whether the foetus has chromosomal abnormalities. The aim is to assess foetal health at an early stage of pregnancy.

According to clinical experience, the main types of abnormal foetuses include **Down syndrome**, **Edwards syndrome**, and **Patau syndrome**, which can be diagnosed respectively by detecting abnormalities in the proportions of cell-free DNA fragments (hereinafter referred to as *chromosomal concentrations*) from foetal chromosomes **21**, **18**, and **13**.

The accuracy of NIPT is primarily determined by the concentration of foetal sex chromosomes (XY for male foetuses, XX for female foetuses). In general, foetal sex chromosome concentrations can be tested between **10 and 25 weeks of gestation**. If the Y-chromosome concentration of a male foetus reaches or exceeds 4%, or if no abnormality is observed in the X-chromosome concentration of a female foetus, the NIPT result is considered to be basically reliable. Otherwise, the accuracy of the test cannot be guaranteed.

In practice, unhealthy foetuses should be identified as early as possible. Delayed detection may shorten the available treatment window:

- **Early detection** (within 12 weeks of gestation) carries relatively low risk;
- **Mid-term detection** (13–27 weeks of gestation) carries high risk;
- **Late detection** (after 28 weeks of gestation) carries extremely high risk.

Empirical evidence shows that the Y-chromosome concentration of male foetuses is closely related to the **gestational age** and the mother's **Body Mass Index (BMI)**. Typically, pregnant women are grouped according to their BMI values (for example: [20, 28), [28, 32), [32, 36), [36, 40), and ≥ 40), and the timing of NIPT (relative to gestational age) is determined for each group.

However, due to individual differences among pregnant women (such as age, BMI, and pregnancy conditions), using simple empirical groupings and a uniform testing time for all women may significantly affect test accuracy. Therefore, rational BMI-based grouping and determination of the optimal NIPT timing for different groups can reduce potential risks associated with shortened treatment windows caused by unhealthy foetuses.

To investigate appropriate NIPT timings for different groups of pregnant women and to analyse test accuracy, the attached dataset provides NIPT data from a certain region (mostly

involving women with high BMI). In actual testing, sequencing failures often occur (for example, due to testing too early or other uncertain factors). In addition, to improve reliability, some women need to undergo multiple blood samplings and tests, or multiple tests from a single blood sample.

Using the provided dataset, establish mathematical models to address the following problems:

- 1) Analyse the correlation between foetal Y-chromosome concentration and factors such as gestational age and maternal BMI. Propose an appropriate relational model and test its statistical significance.
- 2) Clinical evidence shows that maternal BMI is the primary factor affecting the **earliest time at which the Y-chromosome concentration of male fetuses reaches the threshold** (i.e. reaches or exceeds 4%).

Group male-foetus pregnancies reasonably according to BMI, specify the BMI interval and the optimal NIPT timing for each group so as to minimise potential maternal risk, and analyse the impact of measurement error on the results.

- 3) The time at which the Y-chromosome concentration of male fetuses reaches the threshold (4%) is influenced by multiple factors (such as height, weight, and age of the mother). Taking these factors, measurement errors, and the proportion of male fetuses meeting the Y-chromosome concentration threshold into account, determine reasonable BMI groupings and the optimal NIPT timing for each group so as to minimise potential maternal risk, and analyse the impact of measurement error on the results.
- 4) As neither pregnant women nor female fetuses carry Y chromosomes, the key issue is how to determine abnormalities in female fetuses. Using aneuploidy of chromosomes **21**, **18**, and **13** of female fetuses (column AB) as the diagnostic outcome, and comprehensively considering factors such as Z-scores of the X chromosome and the above chromosomes, GC content, read counts and relevant ratios, and BMI, propose a method for determining abnormalities in female fetuses.

Appendix 1: Description of Data Columns in the Dataset

Column	Description	Column	Description
A	Sample ID	Q	Z-score for chromosome 13
B	Pregnant woman ID	R	Z-score for chromosome 18
C	Maternal age	S	Z-score for chromosome 21
D	Maternal height	T	Z-score for chromosome X
E	Maternal weight	U	Z-score for chromosome Y (blank for female fetuses)
F	Date of last menstrual period	V	Y-chromosome concentration (proportion of Y-chromosome cell-free DNA; blank for female fetuses)
G	Method of achieving pregnancy	W	X-chromosome concentration (estimated by bioinformatic analysis under certain assumptions; may be negative)
H	Test date	X	GC content of chromosome 13
I	Number of blood draws	Y	GC content of chromosome 18
J	Gestational age at test (weeks + days)	Z	GC content of chromosome 21
K	Maternal BMI	AA	Proportion of filtered reads to total reads
L	Total raw sequencing reads	AB	Detected aneuploidy of chromosomes 13, 18, or 21 (blank indicates no abnormality)
M	Proportion of reads aligned to reference genome	AC	Number of pregnancies
N	Proportion of duplicate reads	AD	Number of deliveries
O	Number of uniquely aligned reads	AE	Foetal health status (post-birth outcome)
P	GC content: proportion of guanine (G) and cytosine (C) bases; an important sequencing quality indicator. Normal range: 40%–60%. Abnormally high, low, or uneven GC content may indicate sequencing quality issues.		

Appendix 2: Z-score

The Z-score is calculated as:

$$Z = \frac{X - \mu}{\sigma}$$

where

- X is the relative count proportion of the target chromosome in the test sample,
- μ is the mean proportion of that chromosome in a normal reference population,
- σ is the standard deviation of that proportion in the normal population.

In NIPT, Z-score analysis is commonly used for statistical determination of chromosomal aneuploidy. Chromosomal aneuploidy is typically defined as the presence of one or three copies of a chromosome, whereas the normal state consists of two copies. The number of sequencing reads obtained for each chromosome is proportional to its length.